

The logo of The University of Texas at Dallas is a circular seal. It features a large, stylized 'UTD' in the center. The words 'THE UNIVERSITY OF TEXAS AT DALLAS' are written around the top half of the circle, and 'EST. 1969' is at the bottom. Two stars are positioned on the left and right sides of the circle.

Forward Kinematics: Product of Exponentials Formula

CS 6341 Robotics

Instructor Yu Xiang

The University of Texas at Dallas

Twist of a Rigid Body

A twist compactly represents the instantaneous motion of a rigid body — a combination of angular and linear velocity.

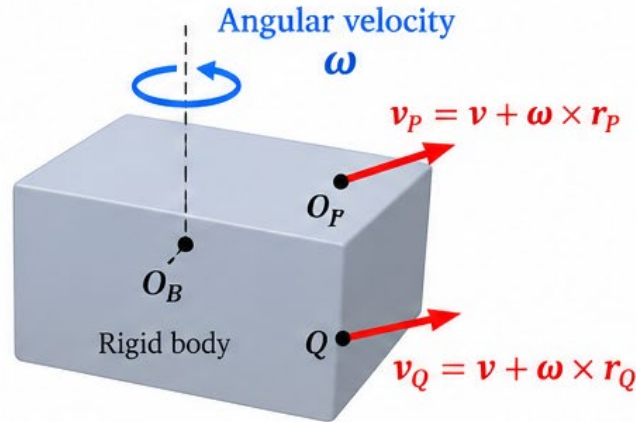
Definition: Twist

A twist is a 6-dimensional vector

$$\mathcal{V} = \begin{bmatrix} \boldsymbol{\omega} \\ \mathbf{v} \end{bmatrix} \in \mathbb{R}^6$$

where

- $\boldsymbol{\omega} \in \mathbb{R}^3$ is the angular velocity (same for all points on the rigid body)
- $\mathbf{v} \in \mathbb{R}^3$ is the linear part of the twist (its meaning depends on the coordinate frame)



Velocity of any point on the rigid body

Given a point with position $\mathbf{r}$ (from the twist reference point), its linear velocity is

$$\mathbf{v}_{\text{point}} = \mathbf{v} + \boldsymbol{\omega} \times \mathbf{r}$$

The angular velocity $\boldsymbol{\omega}$ is the same for all points, while the linear velocity depends on the point location $\mathbf{r}$.

Special cases:

- At the reference point ($\mathbf{r} = \mathbf{0}$): $\mathbf{v}_{\text{point}} = \mathbf{v}$
- At another point: $\mathbf{v}_{\text{point}} = \mathbf{v} + \boldsymbol{\omega} \times \mathbf{r}$

Spatial twist (expressed in $\{S\}$)

The spatial twist is

$$\mathcal{V}^s = \begin{bmatrix} \boldsymbol{\omega}^s \\ \mathbf{v}^s \end{bmatrix} \in \mathbb{R}^6$$

- Expressed in the space (world) frame $\{S\}$.
- The linear part $\mathbf{v}^s$ is **not** the velocity of the body origin in general.
- For any point with position $\mathbf{x}$ (in $\{S\}$):

$$\dot{\mathbf{v}}(\mathbf{x}) = \mathbf{v}^s + \boldsymbol{\omega}^s \times \mathbf{x}$$

Interpretation: $\dot{\mathbf{v}}^s$ and $\boldsymbol{\omega}^s$ define the velocity field of the rigid body in the space frame.

Body twist (expressed in $\{B\}$)

The body twist is

$$\mathcal{V}^b = \begin{bmatrix} \boldsymbol{\omega}^b \\ \mathbf{v}^b \end{bmatrix} \in \mathbb{R}^6$$

- Expressed in the body frame $\{B\}$ (attached to the rigid body).
- The linear part $\mathbf{v}^b$ is the **velocity** of the body frame origin O_B , expressed in $\{B\}$.
- For any point with position $\mathbf{r}$ (in $\{B\}$):

$$\dot{\mathbf{v}}(\mathbf{r}) = \mathbf{v}^b + \boldsymbol{\omega}^b \times \mathbf{r}$$

Interpretation: $\mathbf{v}^b$ is the velocity of the body origin, and $\boldsymbol{\omega}^b$ is the angular velocity, both expressed in the body frame.

Relation between spatial and body twists

The spatial and body twists represent the same motion and are related by the adjoint transformation:

$$\mathcal{V}^s = \text{Ad}_{T_{SB}} \mathcal{V}^b \quad (\text{and} \quad \mathcal{V}^b = \text{Ad}_{T_{SB}}^{-1} \mathcal{V}^s)$$

For $T_{SB} = \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & \mathbf{1} \end{bmatrix}$ (rotation $\mathbf{R}$ and position $\mathbf{p}$ of $\{B\}$ in $\{S\}$),

$$\text{Ad}_{T_{SB}} = \begin{bmatrix} \mathbf{R} & \mathbf{0} \\ [\mathbf{p}]_{\times} & \mathbf{R} \end{bmatrix}$$

where $[\mathbf{p}]_{\times}$ is the skew-symmetric matrix of $\mathbf{p}$.

Twist

- Let's combine angular velocity and linear velocity into a 6D vector called twist

$$\mathcal{V} = \begin{bmatrix} \omega \\ v \end{bmatrix}$$

- Twist can be defined in fixed frame or body frame

Spatial twist $\mathcal{V}_s = \begin{bmatrix} \omega_s \\ v_s \end{bmatrix} \in \mathbb{R}^6$

Body twist $\mathcal{V}_b = \begin{bmatrix} \omega_b \\ v_b \end{bmatrix} \in \mathbb{R}^6$

Relationship between Spatial Twist and Body Twist

$$\begin{bmatrix} \omega_s \\ v_s \end{bmatrix} = \begin{bmatrix} R & 0 \\ [p]R & R \end{bmatrix} \begin{bmatrix} \omega_b \\ v_b \end{bmatrix}$$

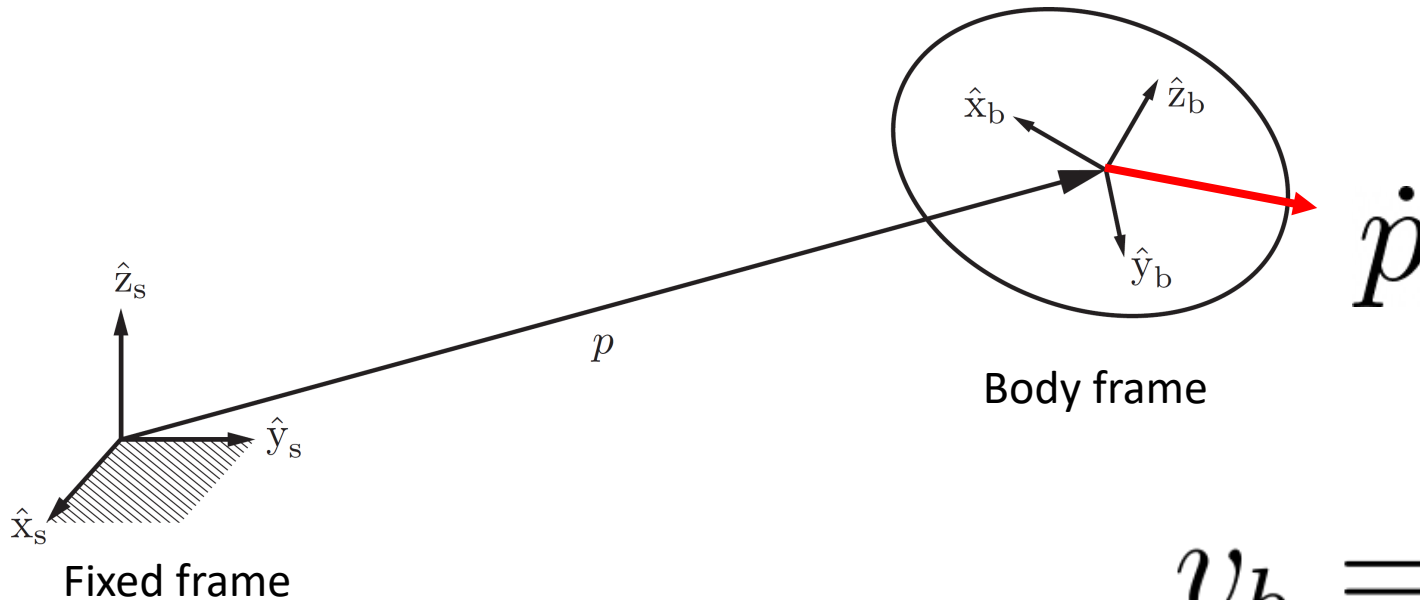


$$[\text{Ad}_T] = \begin{bmatrix} R & 0 \\ [p]R & R \end{bmatrix} \in \mathbb{R}^{6 \times 6}$$

The adjoint representation of $T = (R, p) \in SE(3)$ $\mathcal{V}_s = [\text{Ad}_{T_{sb}}] \mathcal{V}_b$

Relationship between Spatial Twist and Body Twist

- What is the relationship between v_s , v_b and $\dot{p}$? $\dot{T} = \begin{bmatrix} \dot{R} & \dot{p} \\ 0 & 0 \end{bmatrix}$



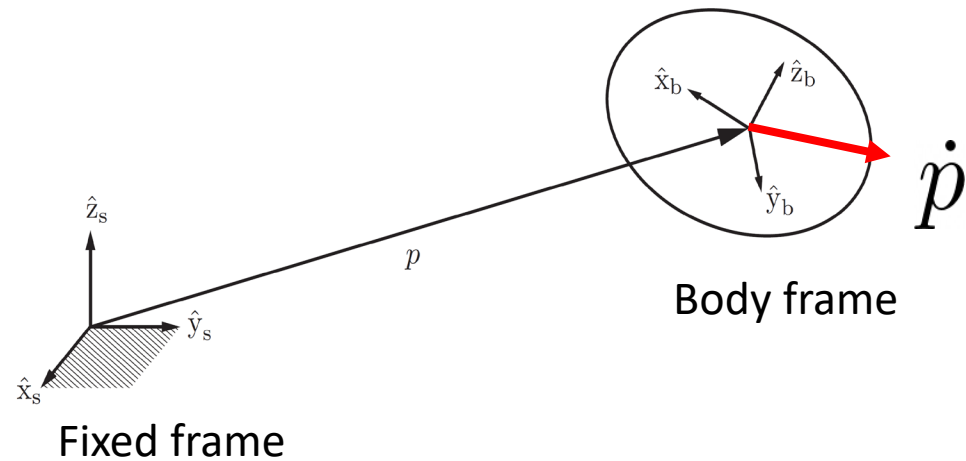
- Linear velocity $\dot{p}$

The linear velocity of the origin of {b} expressed in the fixed frame {s}

$$v_b = R_{bs}\dot{p} = R^T \dot{p}$$

Relationship between Spatial Twist and Body Twist

- What is the relationship between v_s , v_b and $\dot{p}$? $\dot{T} = \begin{bmatrix} \dot{R} & \dot{p} \\ 0 & 0 \end{bmatrix}$



- Linear velocity $\dot{p}$ The linear velocity of the origin of {b} expressed in the fixed frame {s}

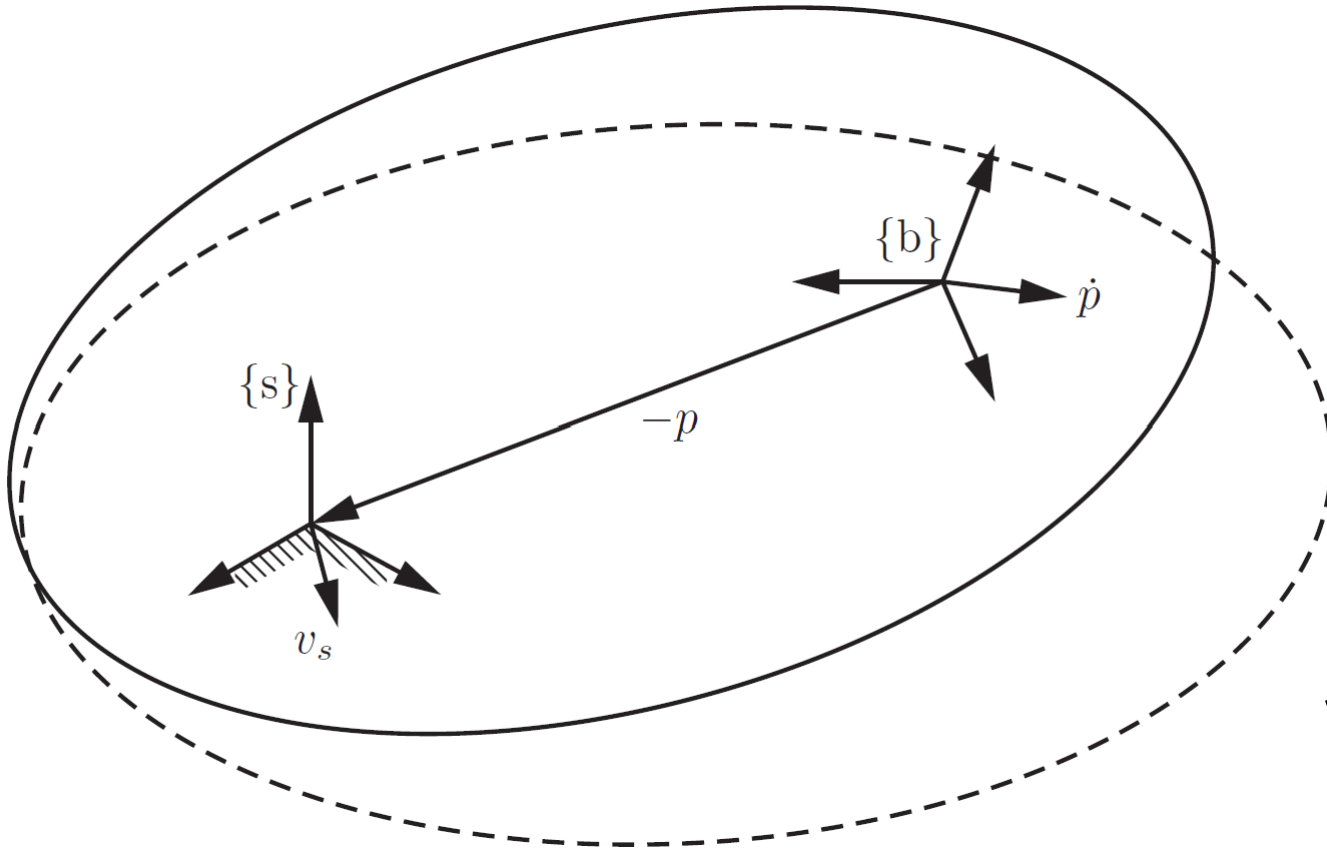
$$v_b = R_{bs}\dot{p} = R^T\dot{p}$$

$$v_s = [p]R\omega_b + Rv_b = [p]\omega_s + RR^T\dot{p}$$

$$v_s = \dot{p} + \omega_s \times (-p) \quad [\omega]p = -[p]\omega$$

Relationship between Spatial Twist and Body Twist

$$v_s = \dot{p} + \omega_s \times (-p)$$



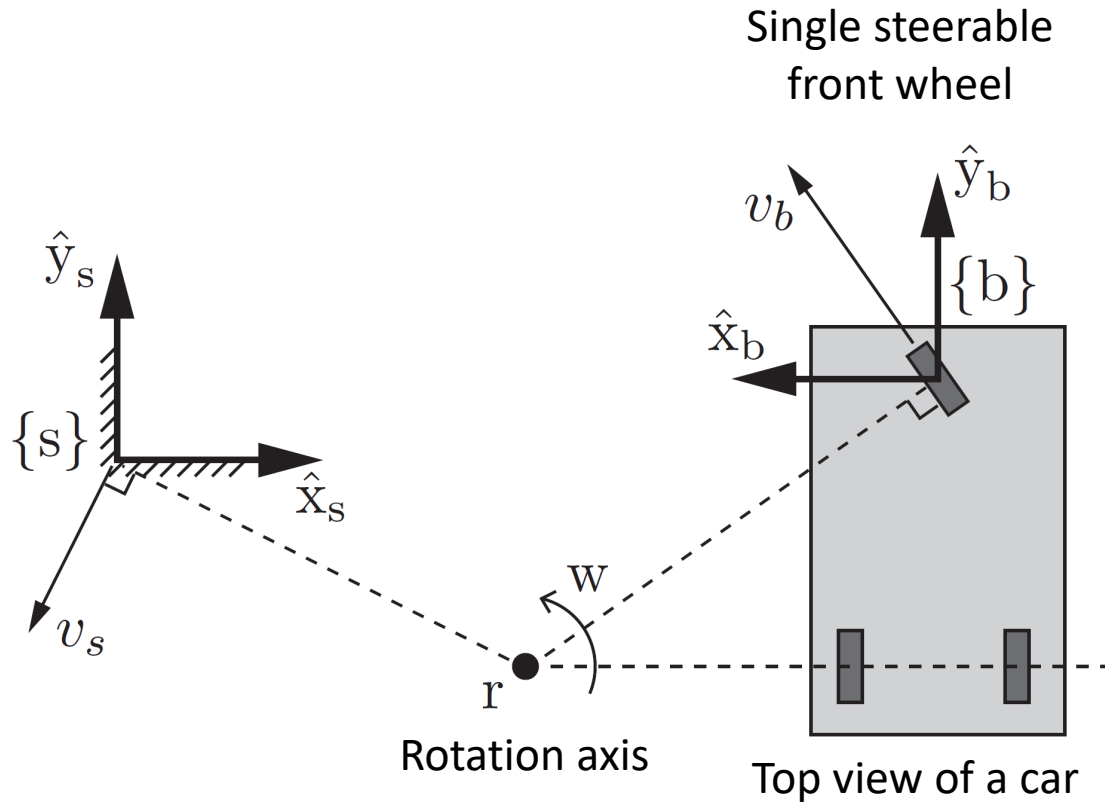
$$\dot{p} = v^s + \omega^s \times p.$$

$$\dot{x} = v^s + \omega^s \times x.$$

$$\dot{x}|_{x=0} = v^s.$$

v_s is the linear velocity that the rigid body's instantaneous velocity field assigns to the origin of the space frame.

Twists Example



• Pure Angular velocity $w = 2 \text{ rad/s}$

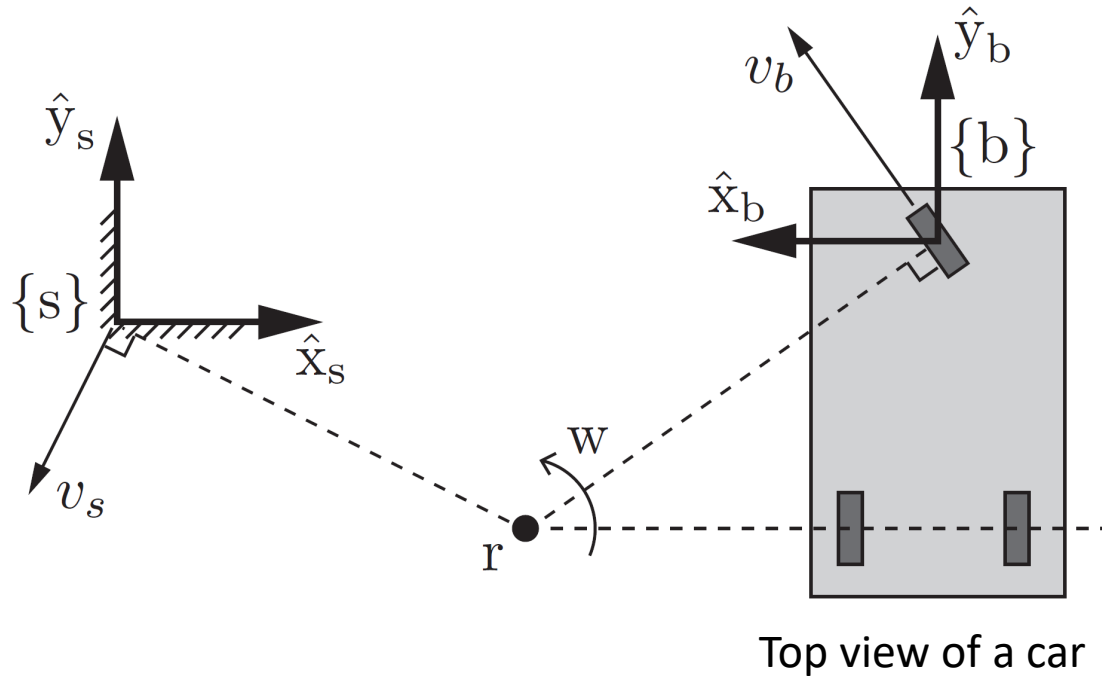
$$r_s = (2, -1, 0) \quad r_b = (2, -1.4, 0)$$

$$\omega_s = (0, 0, 2) \quad \omega_b = (0, 0, -2)$$

$$T_{sb} = \begin{bmatrix} R_{sb} & p_{sb} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 0.4 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

What are the linear velocities? $v_s \quad v_b$

Twists Example



Linear velocity of the car

$$v_s = \omega_s \times (-r_s) = r_s \times \omega_s = (-2, -4, 0),$$

$$v_b = \omega_b \times (-r_b) = r_b \times \omega_b = (2.8, 4, 0),$$

- Pure Angular velocity $\omega = 2 \text{ rad/s}$

$$r_s = (2, -1, 0) \quad r_b = (2, -1.4, 0)$$

$$\omega_s = (0, 0, 2) \quad \omega_b = (0, 0, -2)$$

$$T_{sb} = \begin{bmatrix} R_{sb} & p_{sb} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 0.4 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathcal{V}_s = \begin{bmatrix} \omega_s \\ v_s \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 2 \\ -4 \\ 0 \end{bmatrix}, \quad \mathcal{V}_b = \begin{bmatrix} \omega_b \\ v_b \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -2 \\ 2.8 \\ 0 \end{bmatrix}$$

Cross Product

- Matrix notation

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

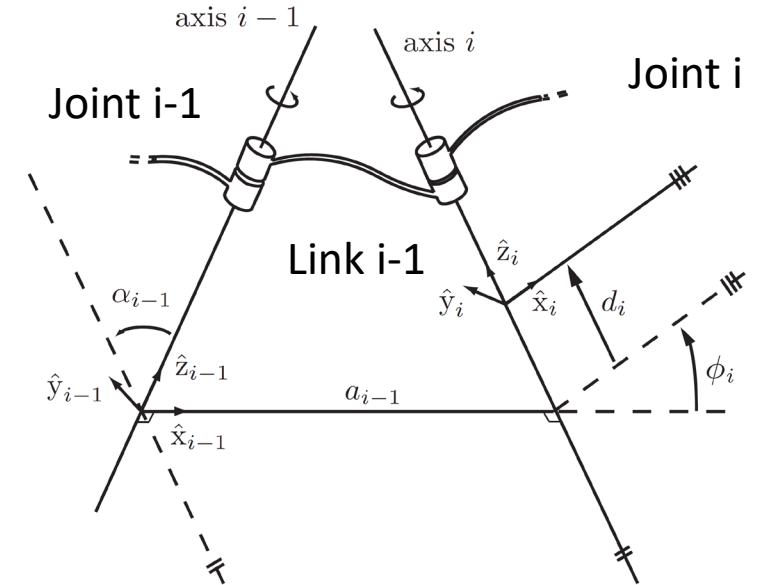
$$\begin{aligned} \mathbf{a} \times \mathbf{b} &= \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \mathbf{k} \\ &= (a_2 b_3 - a_3 b_2) \mathbf{i} - (a_1 b_3 - a_3 b_1) \mathbf{j} + (a_1 b_2 - a_2 b_1) \mathbf{k}. \end{aligned}$$

https://en.wikipedia.org/wiki/Cross_product

Forward Kinematics with D-H Parameters

- Link frame transformation

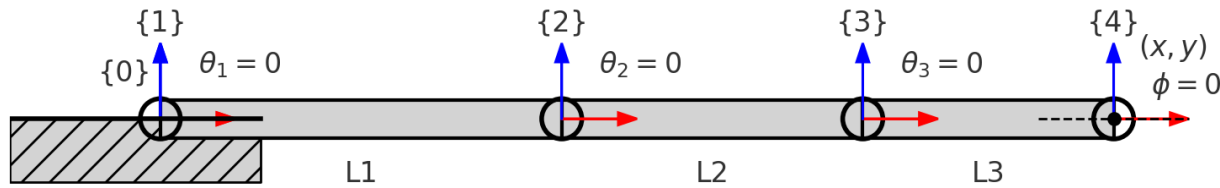
$$T_{i-1,i} = \text{Rot}(\hat{x}, \alpha_{i-1}) \text{Trans}(\hat{x}, a_{i-1}) \text{Trans}(\hat{z}, d_i) \text{Rot}(\hat{z}, \phi_i)$$



$$T_{0n}(\theta_1, \dots, \theta_n) = T_{01}(\theta_1) T_{12}(\theta_2) \cdots T_{n-1,n}(\theta_n)$$

Forward Kinematics: Product of Exponentials Formula

- A different approach
- Define M to the position and orientation of frame {4} when all the joint angles are zeros (“home” or “zero” position of the robot)



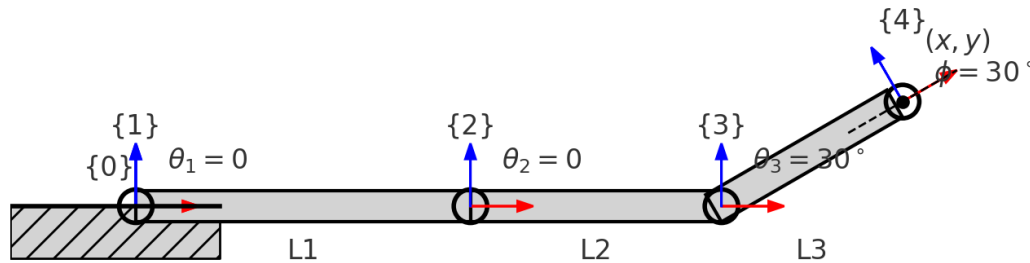
Forward kinematics of a 3R planar open chain.

$$M = \begin{bmatrix} 1 & 0 & 0 & L_1 + L_2 + L_3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Forward Kinematics

- Consider each revolute joint as a zero-pitch (no additional linear velocity) screw-axis expressed in the $\{0\}$ frame (fixed frame)

Planar 3R Arm with $\theta_3 = 30^\circ$, $\theta_1 = \theta_2 = 0$



Forward kinematics of a 3R planar open chain.

$$\mathbf{v}^s = -\boldsymbol{\omega}^s \times \mathbf{q}^s$$

For joint 3

Spatial twist $\mathcal{S}_3 = \begin{bmatrix} \omega_3 \\ v_3 \end{bmatrix}$ $\omega_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

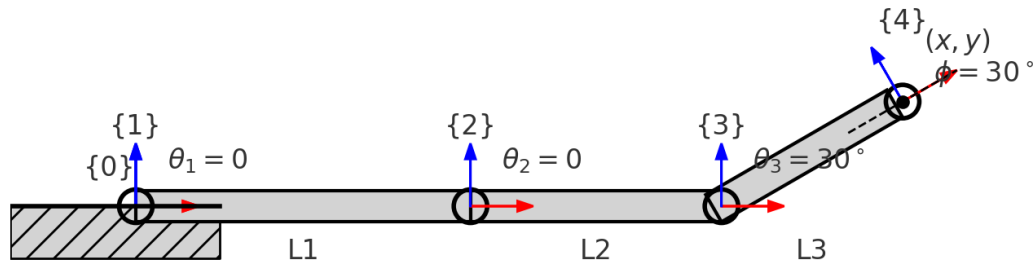
$v_3 = -\omega_3 \times \mathbf{q}_3$ $v_3 = \begin{bmatrix} 0 \\ -(L_1 + L_2) \\ 0 \end{bmatrix}$

$\mathbf{q}_3 = (L_1 + L_2, 0, 0)$

$$\mathcal{S}_3 = \begin{bmatrix} \omega_3 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ -(L_1 + L_2) \\ 0 \end{bmatrix}$$

Forward Kinematics

Planar 3R Arm with $\theta_3 = 30^\circ$, $\theta_1 = \theta_2 = 0$



Forward kinematics of a 3R planar open chain.

- Consider each revolute joint as a zero-pitch (no additional linear velocity) screw-axis expressed in the $\{0\}$ frame (fixed frame)

$$[\mathcal{S}_3] = \begin{bmatrix} [\omega] & v \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & -(L_1 + L_2) \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

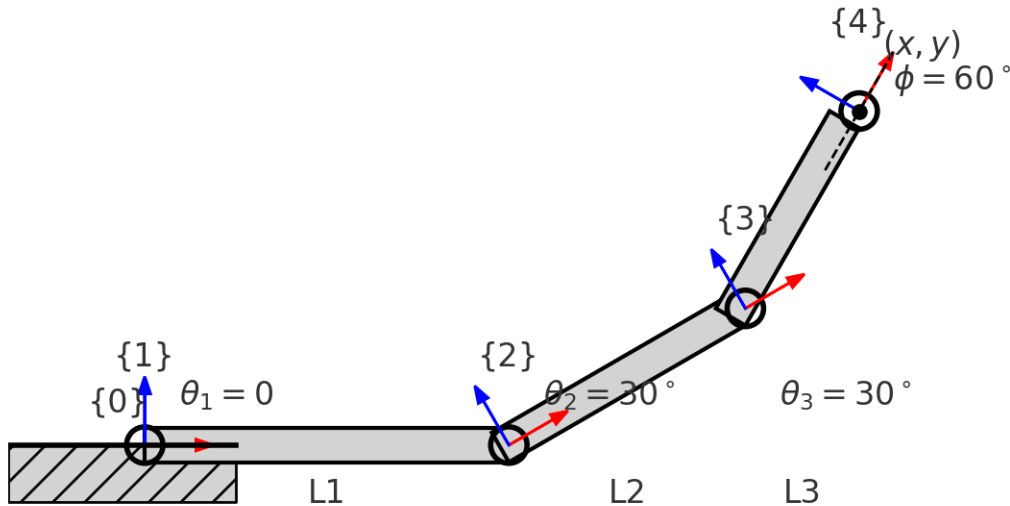
$$T_{04} = e^{[\mathcal{S}_3]\theta_3} M \quad (\text{for } \theta_1 = \theta_2 = 0)$$

$$T(\theta) = e^{[\mathcal{S}]\theta} = \begin{bmatrix} I + \sin \theta [\hat{\omega}] + (1 - \cos \theta) [\hat{\omega}]^2 & (I\theta + (1 - \cos \theta) [\hat{\omega}] + (\theta - \sin \theta) [\hat{\omega}]^2) v \\ 0 & 1 \end{bmatrix}$$

Exponential Coordinates of Rigid-Body Motions

Forward Kinematics

Planar 3R Arm with $\theta_2 = 30^\circ$, $\theta_3 = 30^\circ$, $\theta_1 = 0$



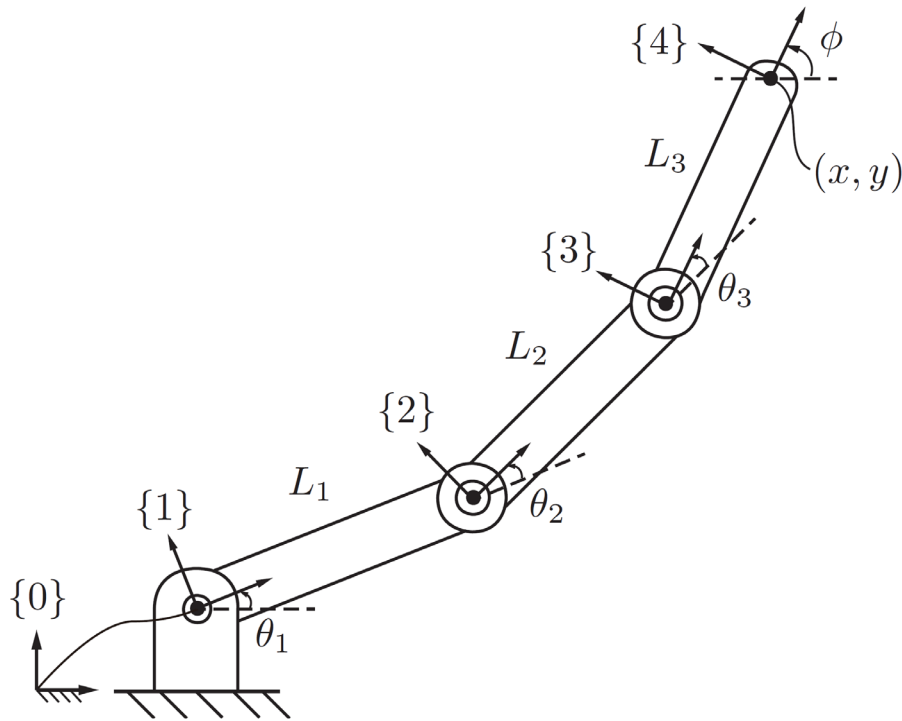
Forward kinematics of a 3R planar open chain.

- Consider each revolute joint as a zero-pitch (no additional linear velocity) screw-axis expressed in the $\{0\}$ frame (fixed frame)

$$[S_2] = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & -L_1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$T_{04} = e^{[S_2]\theta_2} e^{[S_3]\theta_3} M \quad (\text{for } \theta_1 = 0)$$

Forward Kinematics

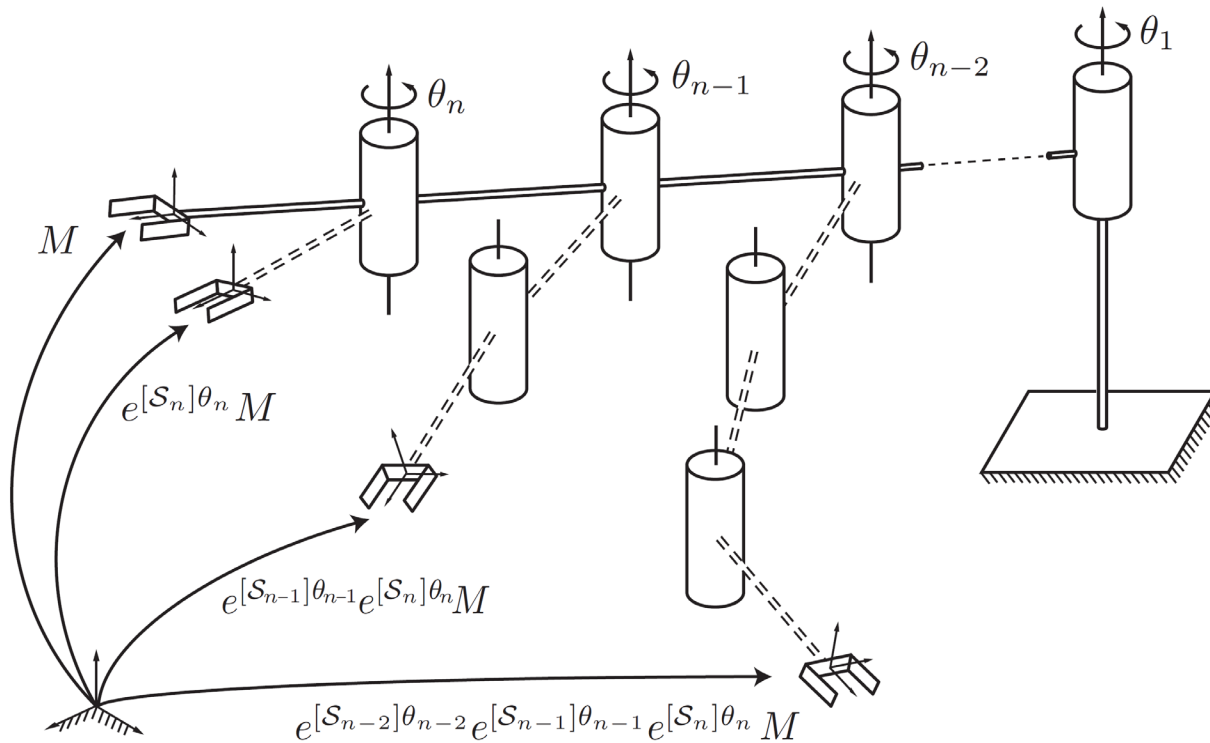


$$[\mathcal{S}_1] = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$T_{04} = e^{[\mathcal{S}_1]\theta_1} e^{[\mathcal{S}_2]\theta_2} e^{[\mathcal{S}_3]\theta_3} M$$

a product of matrix exponentials
(does not use any frame references, only {0} and M)

Product of Exponentials Formula



- Each link apply a screw motion to all the outward links
- Base frame $\{s\}$
- End-effector frame $\{b\}$

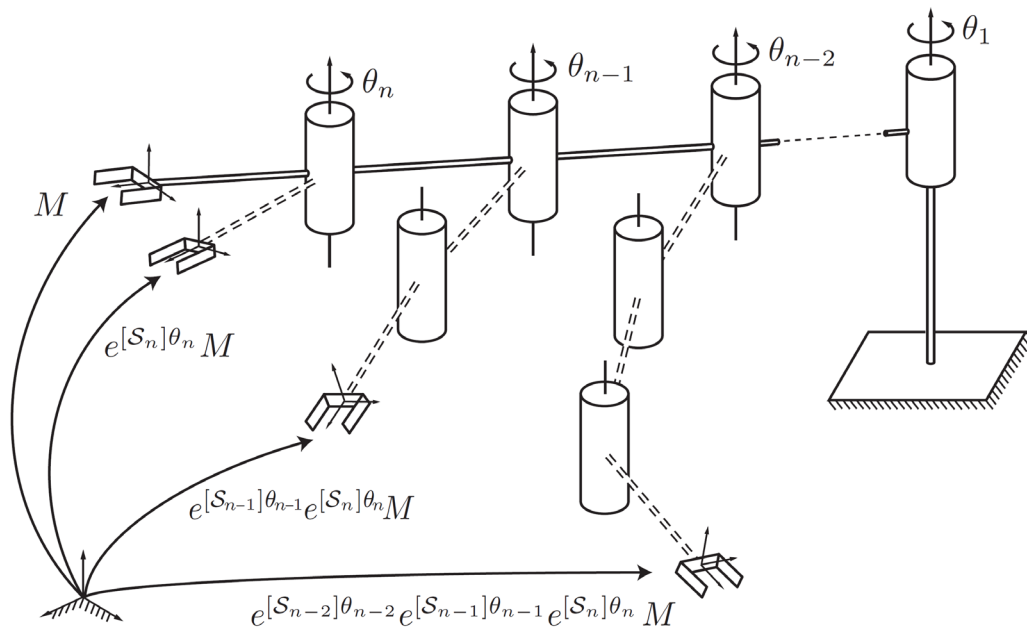
$$M \in SE(3)$$

$\{b\}$ in $\{s\}$ when all the joint values are zeros

$$T = e^{[S_n]\theta_n} M$$

$\{b\}$ in $\{s\}$ when joint n with value θ_n

Product of Exponentials Formula

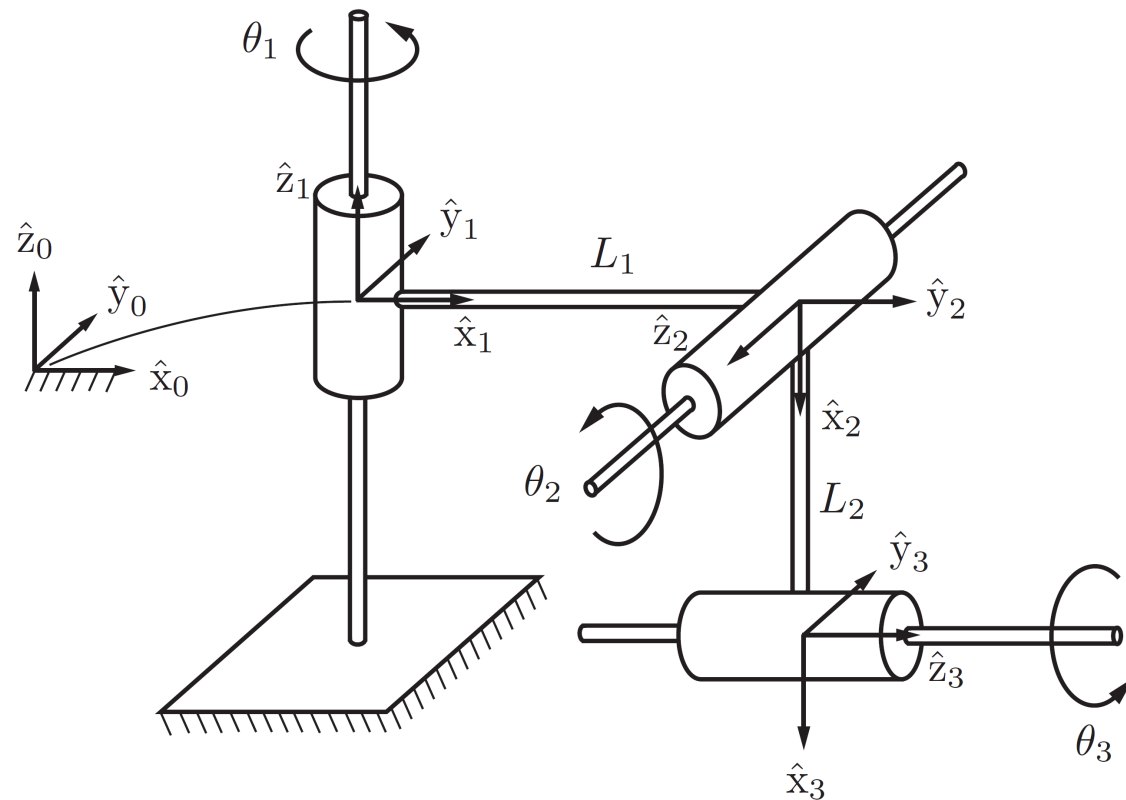


$$T(\theta) = e^{[S_1]\theta_1} \dots e^{[S_{n-1}]\theta_{n-1}} e^{[S_n]\theta_n} M$$

Joint values $(\theta_1, \dots, \theta_n)$

- Space form of the product of exponentials formula
- Unlike D-H representation, no link reference frames need to be defined

Product of Exponentials Formula



A 3R spatial open chain

$$T(\theta) = e^{[\mathcal{S}_1]\theta_1} e^{[\mathcal{S}_2]\theta_2} e^{[\mathcal{S}_3]\theta_3} M$$

$$M = \begin{bmatrix} 0 & 0 & 1 & L_1 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & -L_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathcal{S}_1 = (\omega_1, v_1) \quad \omega_1 = (0, 0, 1) \quad v_1 = (0, 0, 0)$$

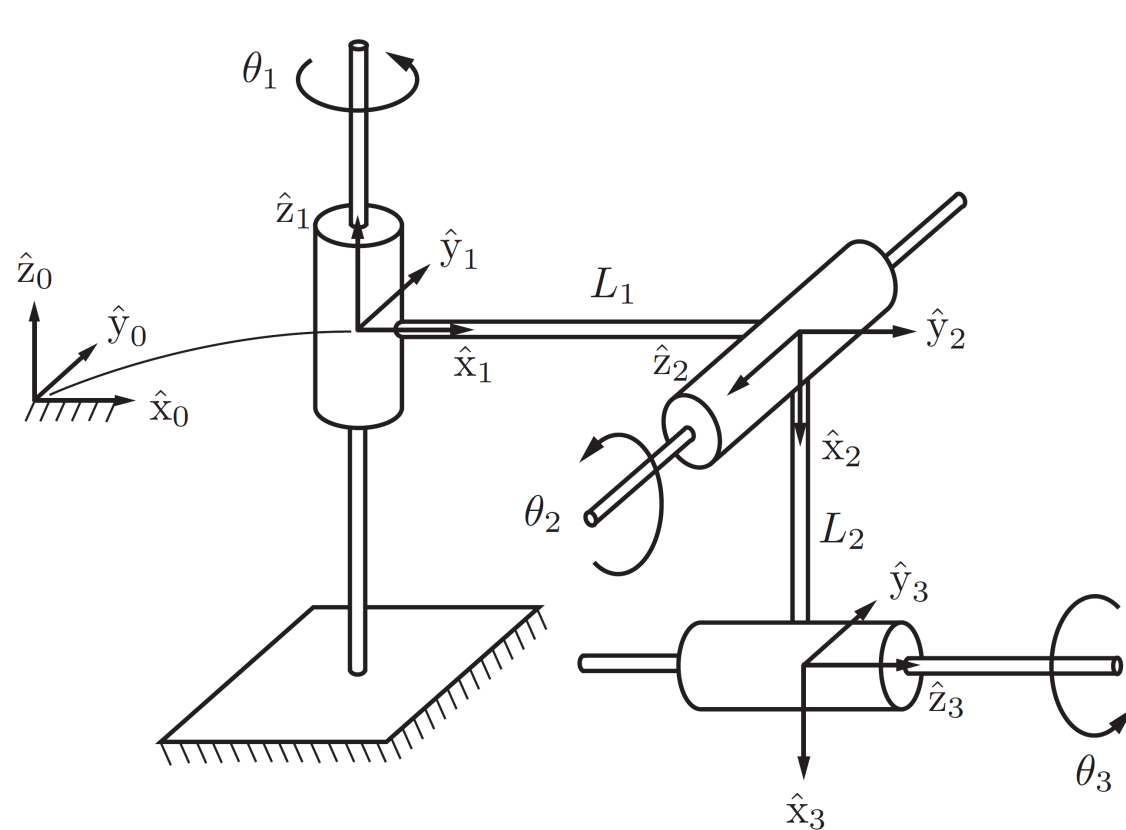
$$\omega_2 = (0, -1, 0) \quad q_2 = (L_1, 0, 0)$$

$$v_2 = -\omega_2 \times q_2 = (0, 0, -L_1)$$

$$\omega_3 = (1, 0, 0) \quad q_3 = (0, 0, -L_2)$$

$$v_3 = -\omega_3 \times q_3 = (0, -L_2, 0)$$

Product of Exponentials Formula



A 3R spatial open chain

$$[S_1] = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$[S_2] = \begin{bmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -L_1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$[S_3] = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & -L_2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

i	ω_i	v_i
1	$(0, 0, 1)$	$(0, 0, 0)$
2	$(0, -1, 0)$	$(0, 0, -L_1)$
3	$(1, 0, 0)$	$(0, L_2, 0)$

Can we use body twist?

$$\mathcal{V}_s = [\text{Ad}_{T_{sb}}] \mathcal{V}_b \quad [\text{Ad}_T] = \begin{bmatrix} R & 0 \\ [p]R & R \end{bmatrix} \in \mathbb{R}^{6 \times 6}$$

Additional results about twist (Lynch & Park 3.3.2)

$$\dot{R}R^{-1} = [\omega_s]$$

$$R^{-1}\dot{R} = [\omega_b]$$

$$[\mathcal{V}_s] = \begin{bmatrix} [\omega_s] & v_s \\ 0 & 0 \end{bmatrix} = \dot{T}T^{-1} \in se(3)$$

$$\begin{aligned} [\mathcal{V}_b] &= T^{-1}\dot{T} \\ &= T^{-1}[\mathcal{V}_s]T \end{aligned}$$

$$[\mathcal{V}_b] = \begin{bmatrix} [\omega_b] & v_b \\ 0 & 0 \end{bmatrix} = T^{-1}\dot{T}$$

$$[\mathcal{V}_s] = T[\mathcal{V}_b]T^{-1}$$

Screw Axes in the End-Effector Frame

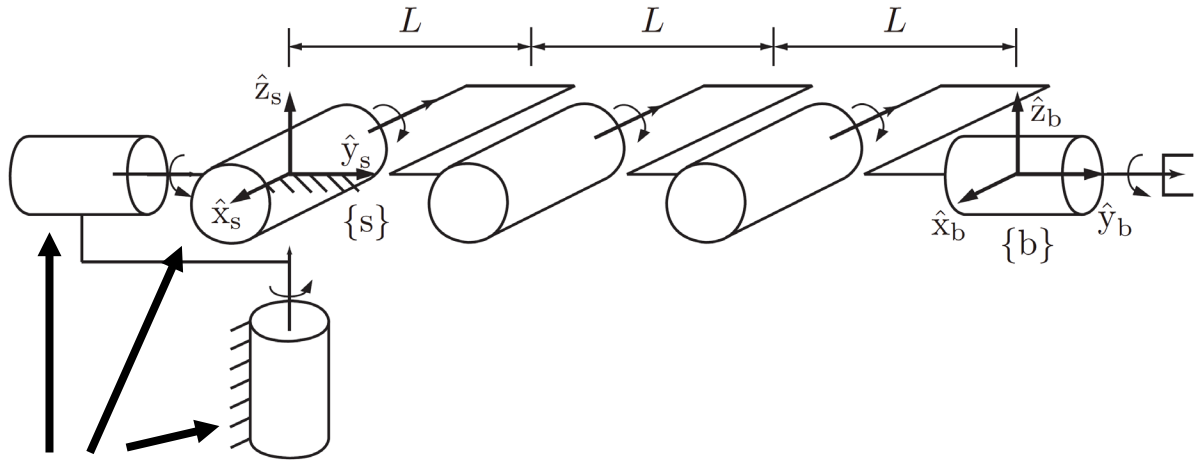
- Proposition 3.10 (Lynch & Park) $e^{M^{-1}PM} = M^{-1}e^PM$ $Me^{M^{-1}PM} = e^PM$
- PoE formula

$$\begin{aligned}
 T(\theta) &= e^{[S_1]\theta_1} \dots \boxed{e^{[S_n]\theta_n} M} & [\mathcal{B}_i] &= M^{-1}[S_i]M \\
 &= e^{[S_1]\theta_1} \dots Me^{M^{-1}[S_n]M\theta_n} & \mathcal{B}_i &= [\text{Ad}_{M^{-1}}]S_i, \quad i = 1, \dots, n \\
 &= e^{[S_1]\theta_1} \dots Me^{M^{-1}[S_{n-1}]M\theta_{n-1}} e^{M^{-1}[S_n]M\theta_n} \\
 &= Me^{M^{-1}[S_1]M\theta_1} \dots e^{M^{-1}[S_{n-1}]M\theta_{n-1}} e^{M^{-1}[S_n]M\theta_n} \\
 &= Me^{[\mathcal{B}_1]\theta_1} \dots e^{[\mathcal{B}_{n-1}]\theta_{n-1}} e^{[\mathcal{B}_n]\theta_n}
 \end{aligned}$$

Body form of the product of exponentials formula

Screw axes $\mathcal{B}_i$ in the end-effector (body) frame when the robot is at its zero position

Screw Axes in the End-Effector Frame



These 3 joints
are at the same
location

PoE forward kinematics for the 6R open chain

$$v^s = -\omega^s \times q^s$$

$$v^b = -\omega^b \times q^b$$

$$M = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 3L \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

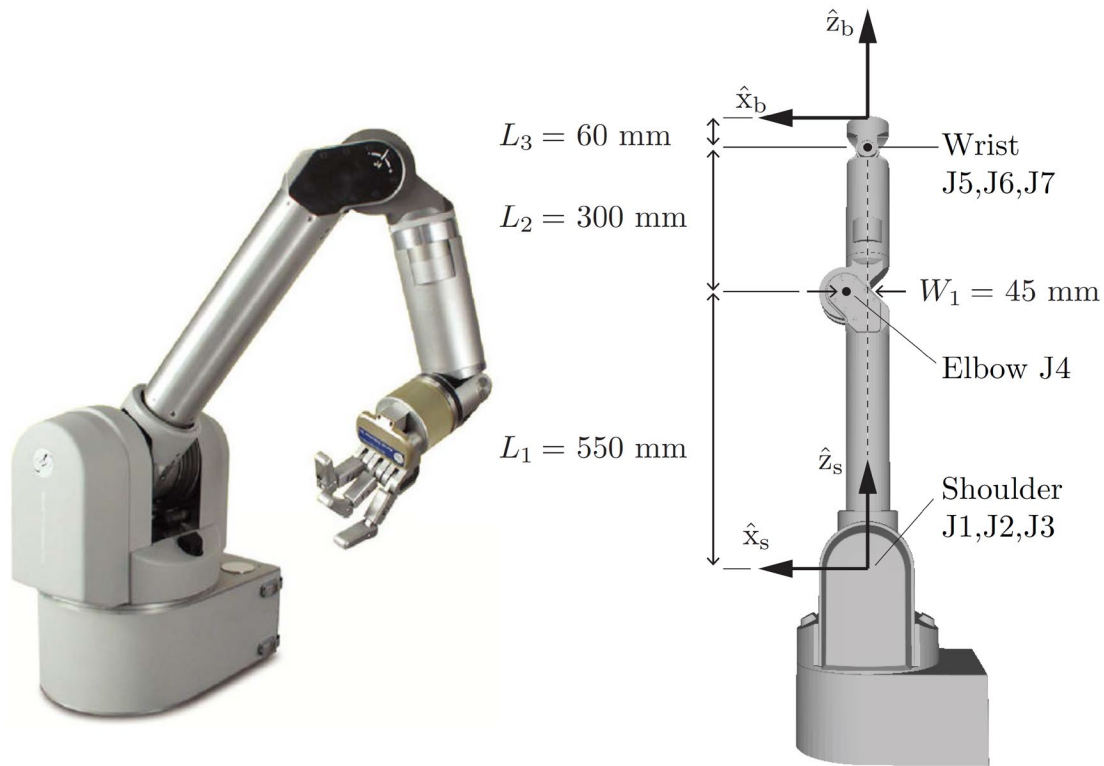
i	ω_i	v_i
1	$(0, 0, 1)$	$(0, 0, 0)$
2	$(0, 1, 0)$	$(0, 0, 0)$
3	$(-1, 0, 0)$	$(0, 0, 0)$
4	$(-1, 0, 0)$	$(0, 0, L)$
5	$(-1, 0, 0)$	$(0, 0, 2L)$
6	$(0, 1, 0)$	$(0, 0, 0)$

Space form

i	ω_i	v_i
1	$(0, 0, 1)$	$(-3L, 0, 0)$
2	$(0, 1, 0)$	$(0, 0, 0)$
3	$(-1, 0, 0)$	$(0, 0, -3L)$
4	$(-1, 0, 0)$	$(0, 0, -2L)$
5	$(-1, 0, 0)$	$(0, 0, -L)$
6	$(0, 1, 0)$	$(0, 0, 0)$

Body form

Screw Axes in the End-Effector Frame



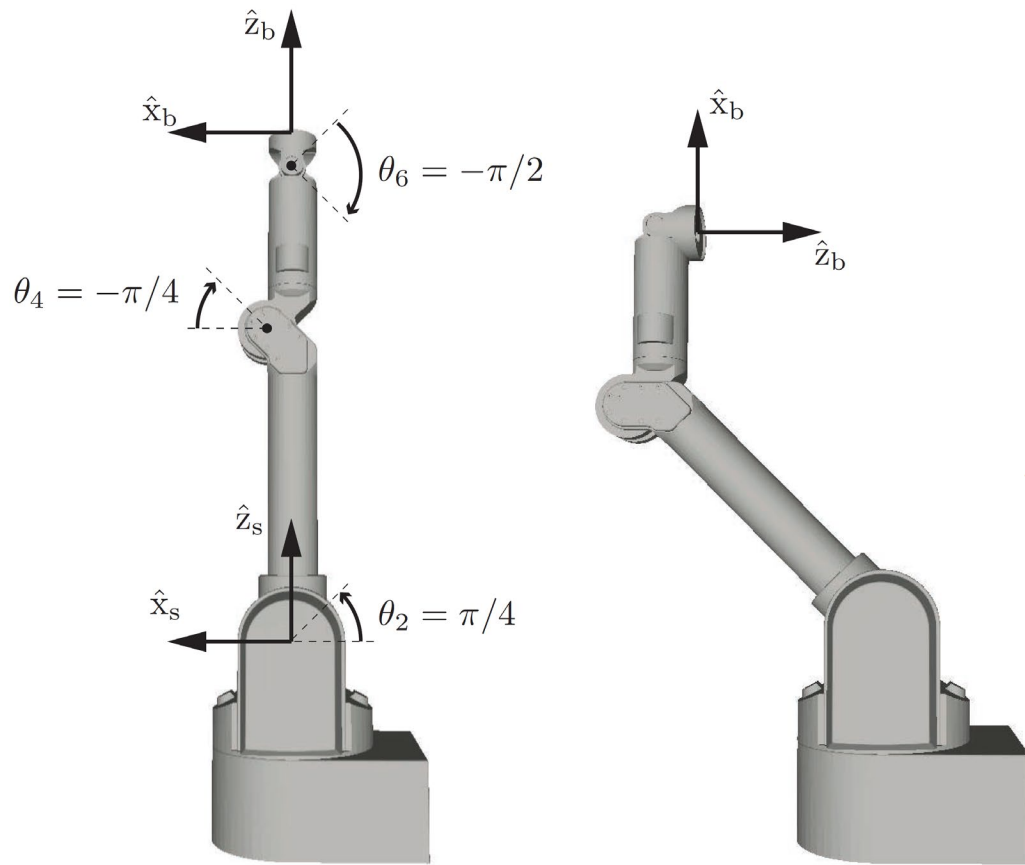
Barrett Technology's WAM 7R robot arm at its zero configuration

$$M = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & L_1 + L_2 + L_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathcal{B}_i = (\omega_i, v_i)$$

i	ω_i	v_i
1	(0, 0, 1)	(0, 0, 0)
2	(0, 1, 0)	$(L_1 + L_2 + L_3, 0, 0)$
3	(0, 0, 1)	(0, 0, 0)
4	(0, 1, 0)	$(L_2 + L_3, 0, W_1)$
5	(0, 0, 1)	(0, 0, 0)
6	(0, 1, 0)	$(L_3, 0, 0)$
7	(0, 0, 1)	(0, 0, 0)

Screw Axes in the End-Effector Frame



$$\theta_2 = 45^\circ, \theta_4 = -45^\circ, \theta_6 = -90^\circ$$

$$T(\theta) = M e^{[\mathcal{B}_2]\pi/4} e^{-[\mathcal{B}_4]\pi/4} e^{-[\mathcal{B}_6]\pi/2} = \begin{bmatrix} 0 & 0 & -1 & 0.3157 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0.6571 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Barrett Technology's WAM 7R robot arm at its zero configuration

Summary

- Screw axis
- Product of Exponentials Formula Spatial twists
 - Spatial form
 - Body form

Further Reading

- Chapter 3 and Chapter 4 in Kevin M. Lynch and Frank C. Park. Modern Robotics: Mechanics, Planning, and Control. 1st Edition, 2017